

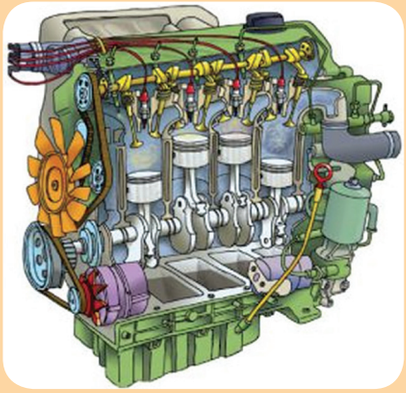
Nikolaus Otto

Internal compression engine

Nikolaus August Otto (14 June 1832, Holzhausenan der Haide, Nassau – 26 January 1891, Cologne) was a German engineer who successfully developed the compressed charge internal combustion engine which ran on petroleum gas and led to the modern internal combustion engine.

The VDI (Association of German Engineers) created DIN standard 1940 which says "Otto Engine: internal combustion engine in which the compressed fuel-air mixture is initiated by a timed spark ignition", which has been applied to all engines of this type since.

His main interest in school had been in science and technology but he graduated after three years as a business apprentice in a small merchandise company.



4.3 TECHNICAL SPECIFICATION OF THE ENGINE

1. Top Dead Centre (TDC) is the outermost point of forward travel of the piston in the cylinder.
2. Bottom Dead Centre (BDC) is the innermost point of backward travel of the piston in the cylinder.
3. Stroke Length is the distance between TDC and BDC travelled by the piston in the cylinder. It will be twice the crankshaft throw.
4. Crankshaft throw is the distance between the centre of the crankshaft main bearing to the centre of the crank pin. It will be half of the stroke length.
5. Cylinder bore is the inside diameter of the cylinder.
6. Clearance volume is the volume of the cylinder above the piston when the piston is at TDC.
7. Swept volume / Displaced volume is the volume displaced by the piston when piston moves from TDC to BDC
8. Total volume is the volume of the cylinder above the piston when the piston is at BDC.

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